Signals and Systems
SET214
Lecture (04)- CT Signals Operations

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Outlines

- Time Shifting
- Time Compression/Expansion
- Exercise
- Amplitude Scaling
- Amplitude Shifting
- Exercise
- Addition/ Multiplication Signals

Outlines

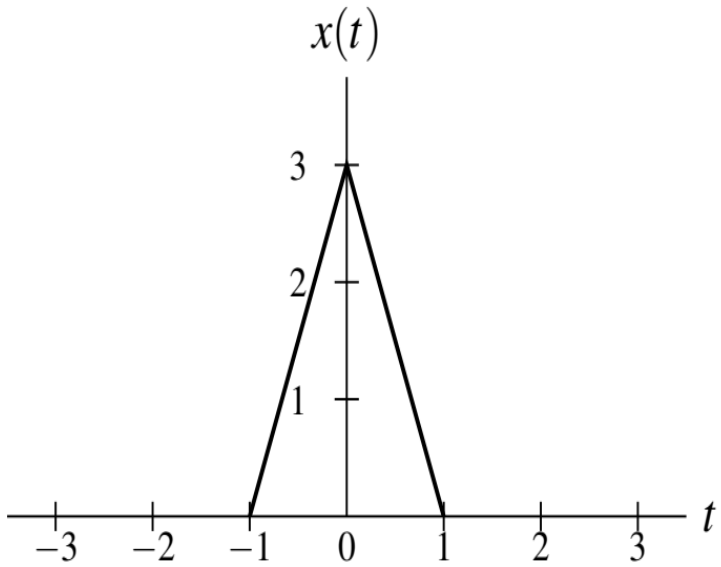
- Time Shifting
- Time Compression/Expansion
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- Amplitude Scaling
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Time Shifting

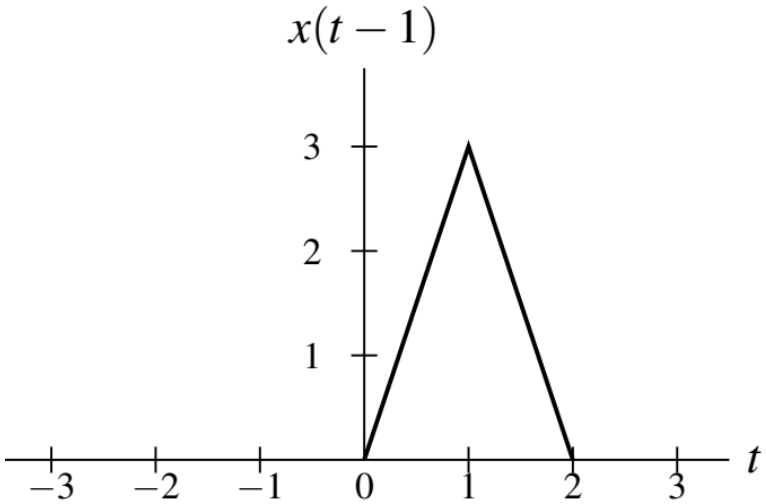
- Time shifting (also called translation) maps the input function x to the output function y as given by:-

  y(t) = x(t -b), where b is a real number

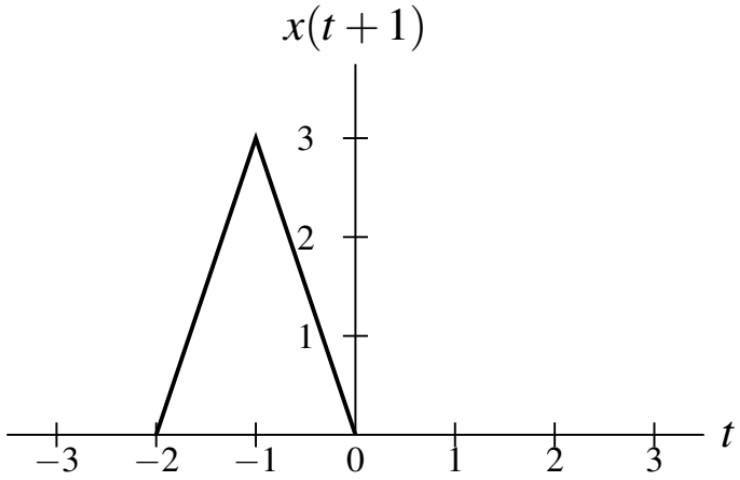
- Such a transformation shifts the function (to the left or right) along the time axis.
  - If b > 0, y is shifted to the right by |b|, relative to x (i.e., delayed in time).
  - If b < 0, y is shifted to the left by |b|, relative to x (i.e., advanced in time).



Original Signal



delayed in time



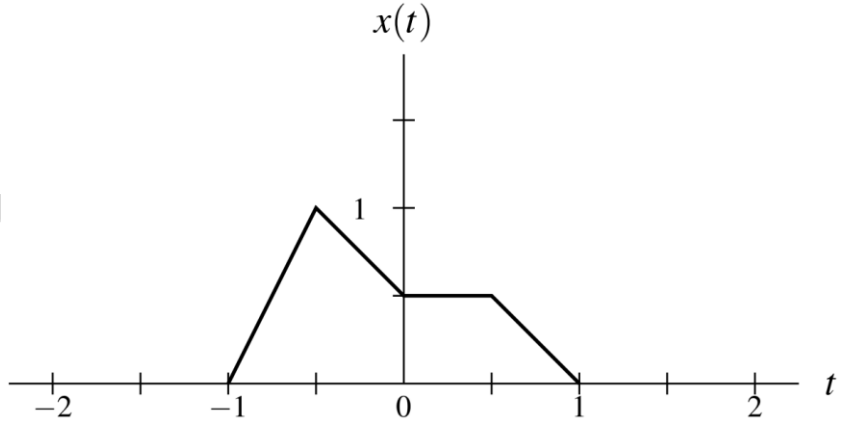
advanced in time

Outlines

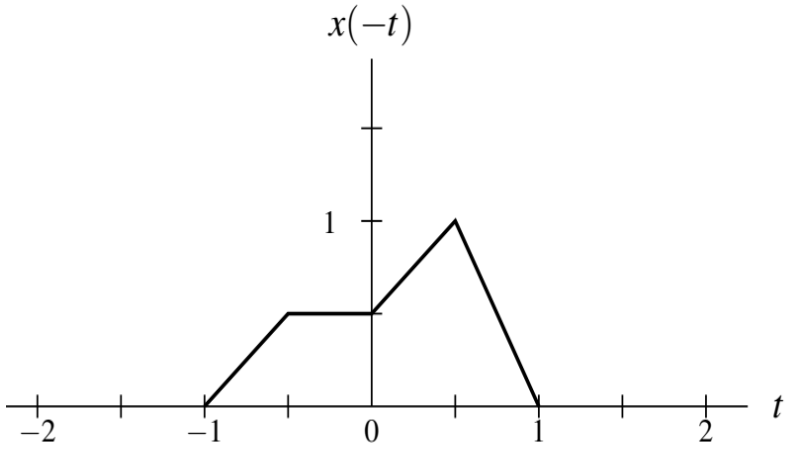
- Time Shifting
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Time Compression/Expansion

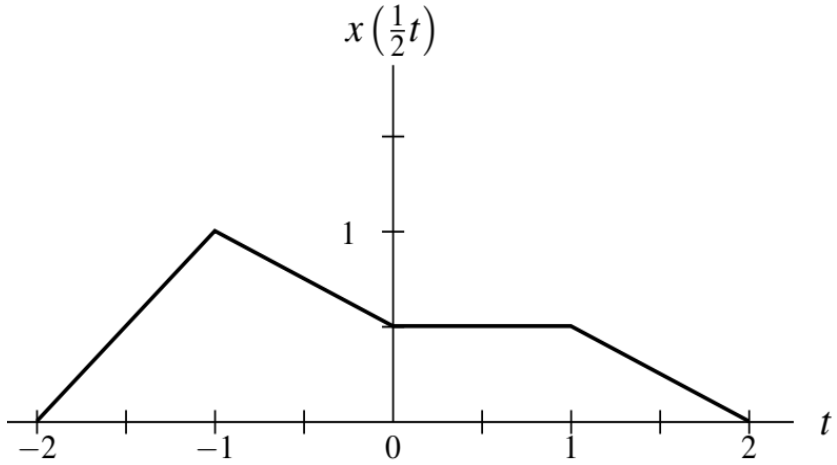
- Time compression/expansion (also called dilation) maps the input function x to the output function y as given by:-
  y(t) = x(at); where *a* is a strictly positive real number
- Such a transformation is associated with a compression/expansion along the time axis.
  - If a > 1, y is compressed along the horizontal axis by a factor of a, relative to x.
  - If a < 1, y is expanded (i.e., stretched) along the horizontal axis by a factor of a , relative to x.
  - If a < 0, the function is also time reversed



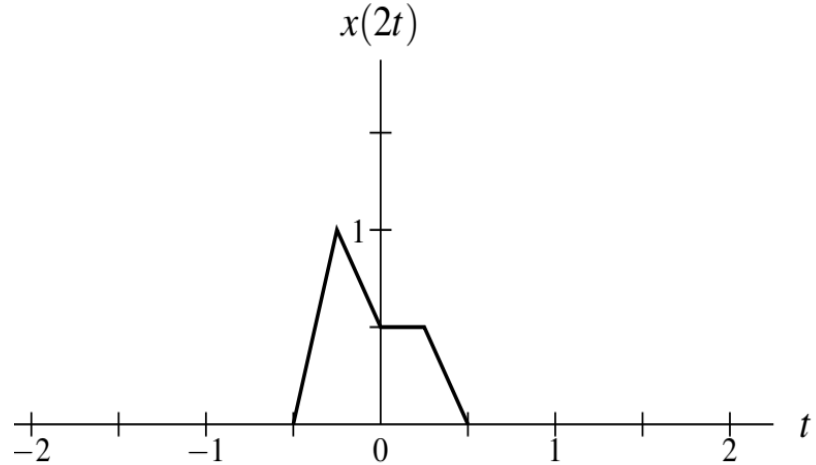
Original Signal



reversed



Expanded



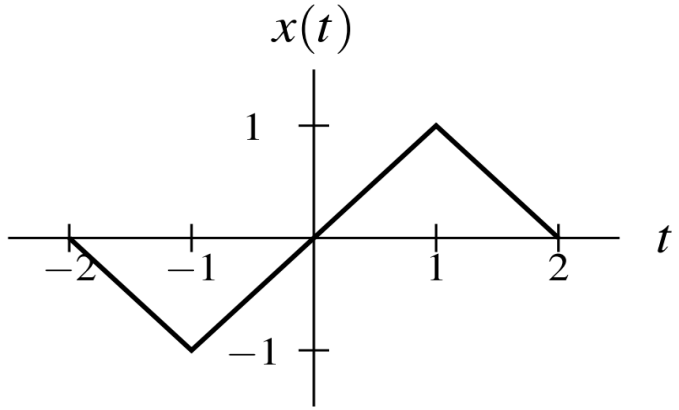
Compressed

Outlines

- Time Shifting
- Time Compression/Expansion
- Exercise
- Amplitude Scaling
- Amplitude Shifting
- Exercise
- Addition/ Multiplication Signals

Exercise

❑ Given x(t) as shown below, plot x(2t -1)

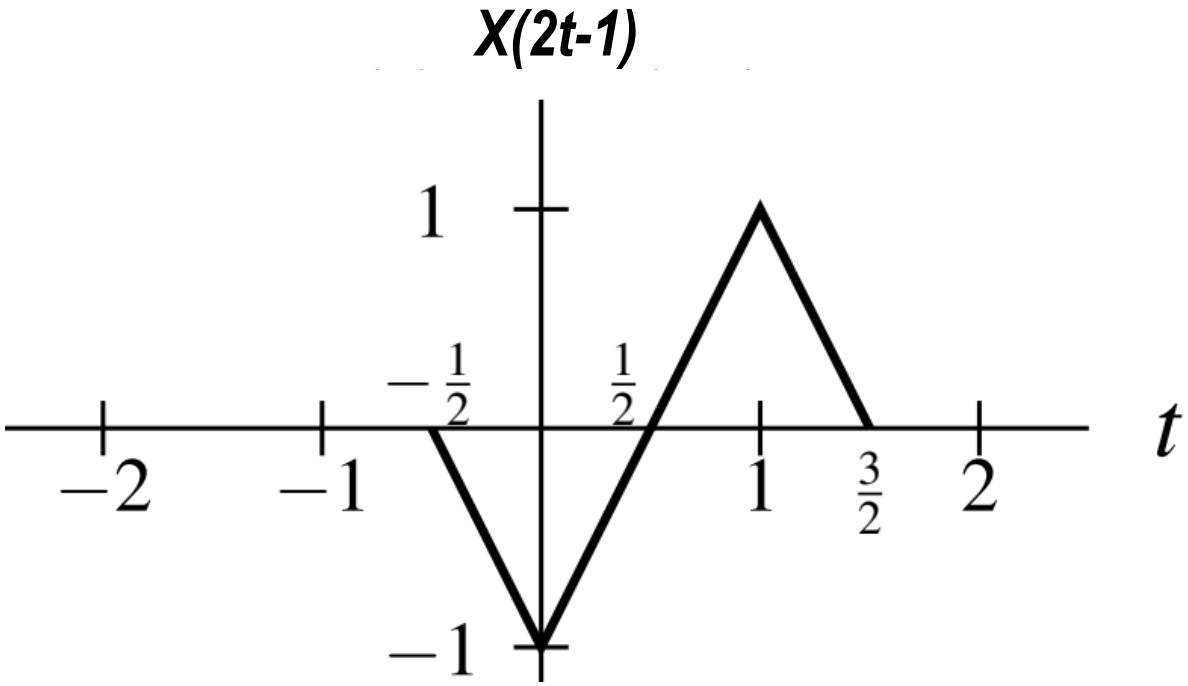


Ans.

❑ Try to build a table as the next

t	X(t)	$t_{new}=(t_{old}+1)/2$	$x(2t-1)$
-2	0	-1/2	0
-1	-1	0	-1
0	0	1/2	0
1	1	1	1
2	0	3/2	0

Notes: Combined Time Scaling and Time Shifting



Outlines

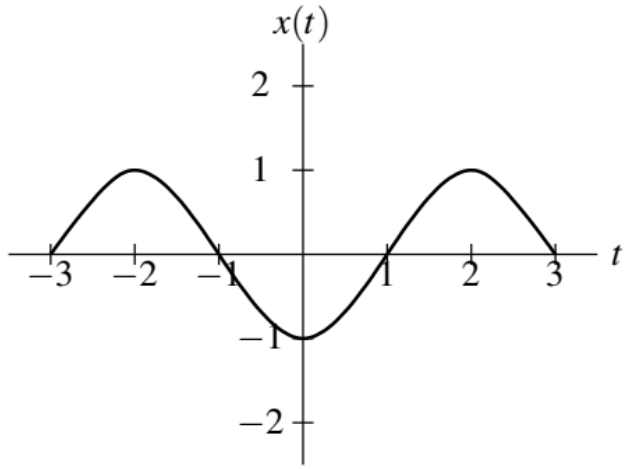
- Time Shifting
- Time Compression/Expansion
- Exercise
- Amplitude Scaling
- Amplitude Shifting
- Exercise
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Amplitude Scaling

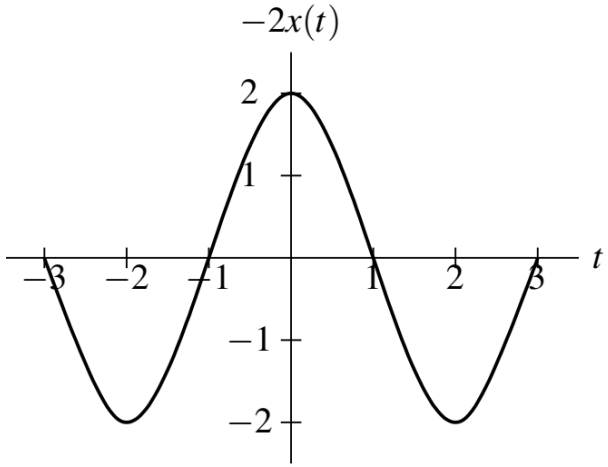
- **Amplitude scaling** maps the input function x to the output function y as given by:-

  $y(t) = ax(t)$, where a is a real number

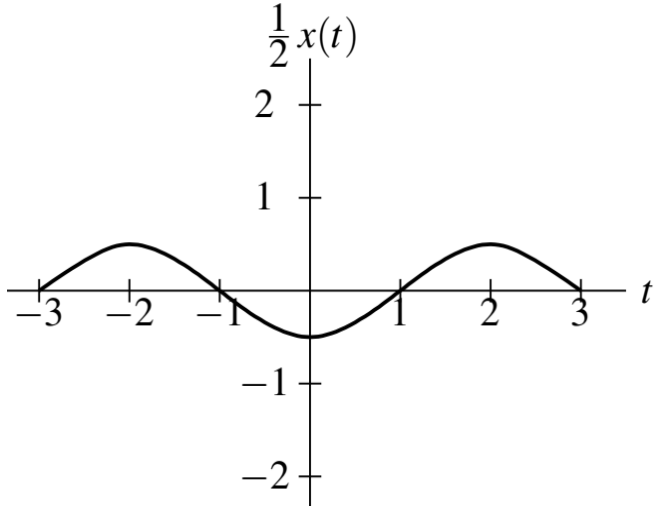
- Geometrically, the output function y is amplified/attenuated in amplitude.
  - If a > 1, y is **amplified** in amplitude by a factor of a
  - If a < 1, y is **attenuated** in amplitude by a factor of a
  - If a < 0, the function is also time reversed



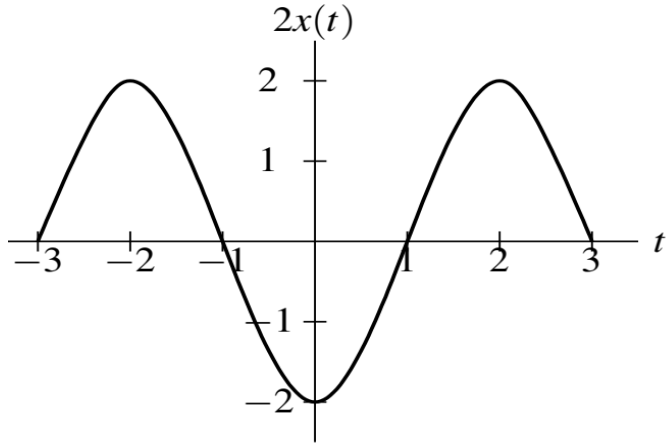
Original Signal



?



Attenuated Signal



Amplified Signal

Outlines

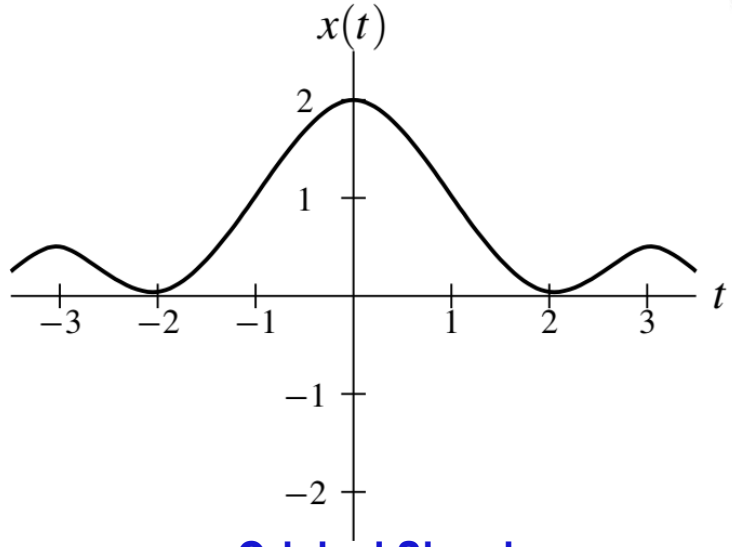
- Time Shifting
- Time Compression/Expansion
- Exercise
- Amplitude Scaling
- Amplitude Shifting
- Exercise
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Amplitude Shifting

- **Amplitude shifting** maps the input function *x* to the output function *y* as given by:-

$y(t) = x(t) + b$, where b is a real number

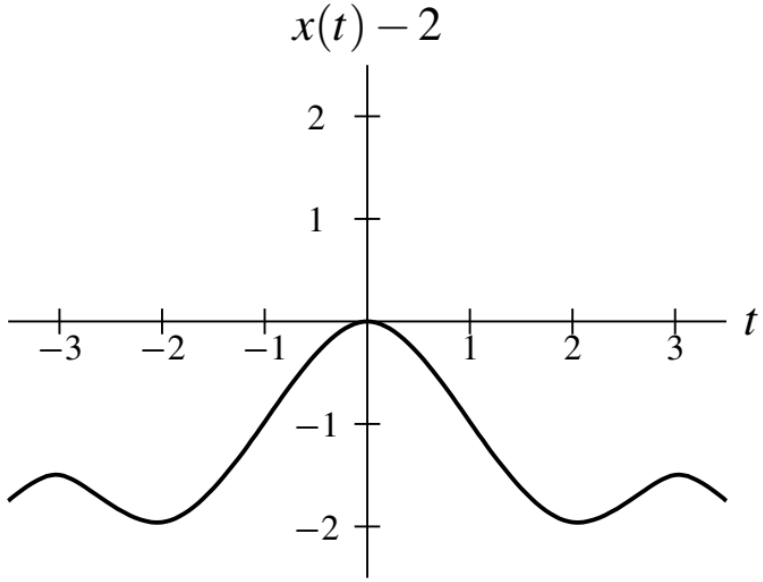
- Geometrically, amplitude shifting adds a *vertical displacement* to *x*.



Original Signal

X(t)+2

?



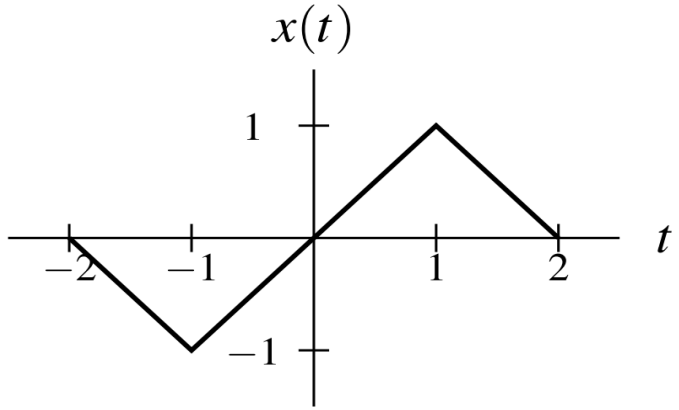
Shifted Signal

Outlines

- Time Shifting
- Time Compression/Expansion
- Exercise
- Amplitude Scaling
- Amplitude Shifting
- Exercise
- Addition/ Multiplication Signals

Exercise

❑ Given x(t) as shown below, plot 2x(t)+1

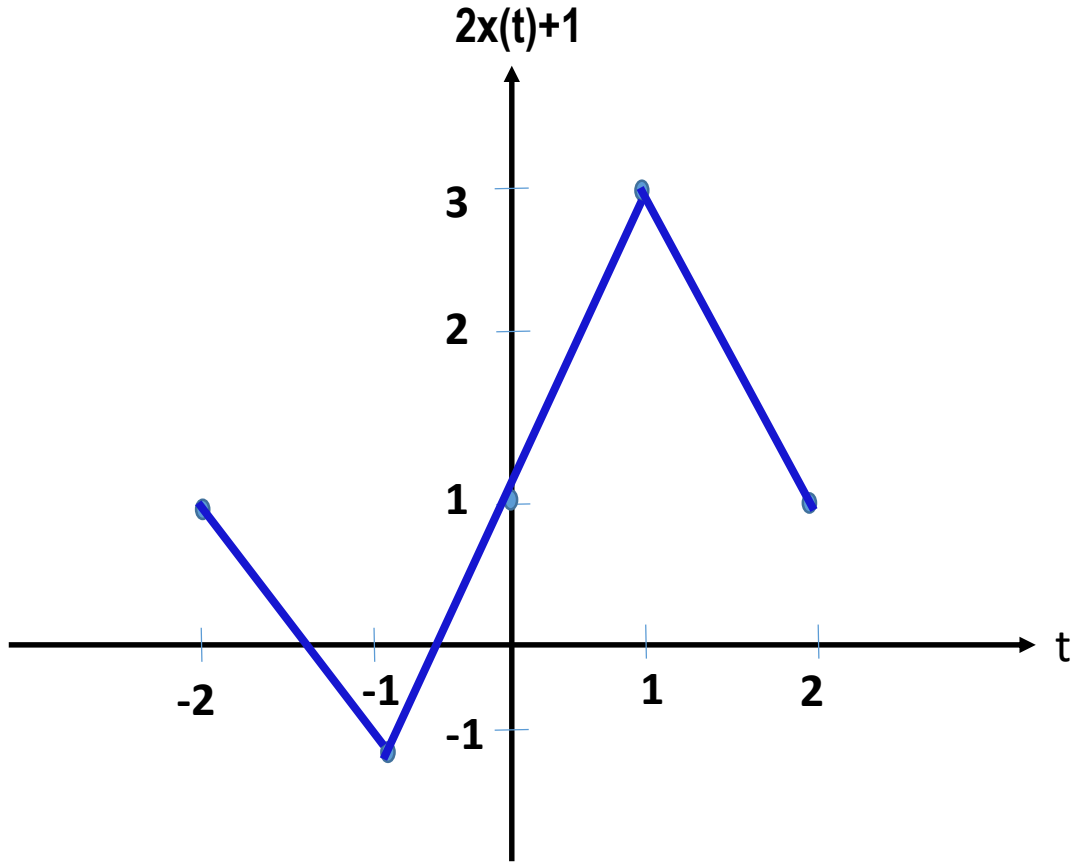


Ans.

❑ Try to build a table as the next

t	X(t)	2x(t)	2x(t)+1
-2	0	0	1
-1	-1	-2	-1
0	0	0	1
1	1	2	3
2	0	0	1

Notes: Combined amplitude Scaling and amplitude Shifting

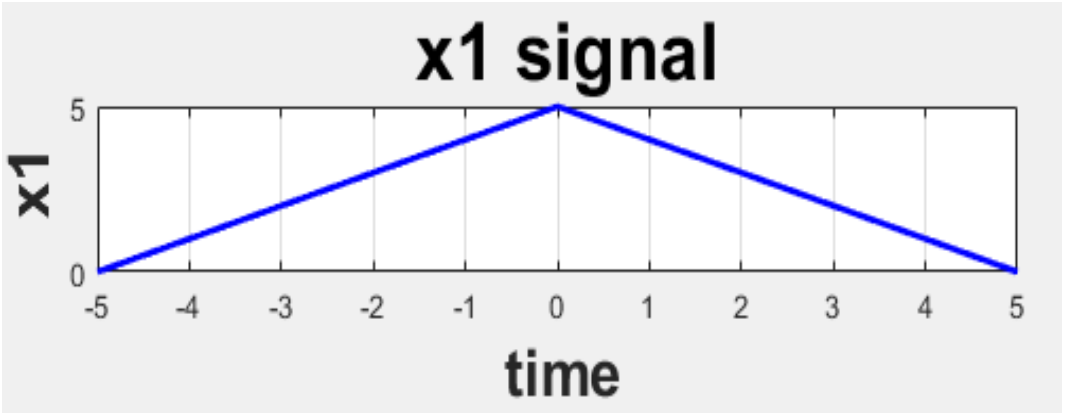


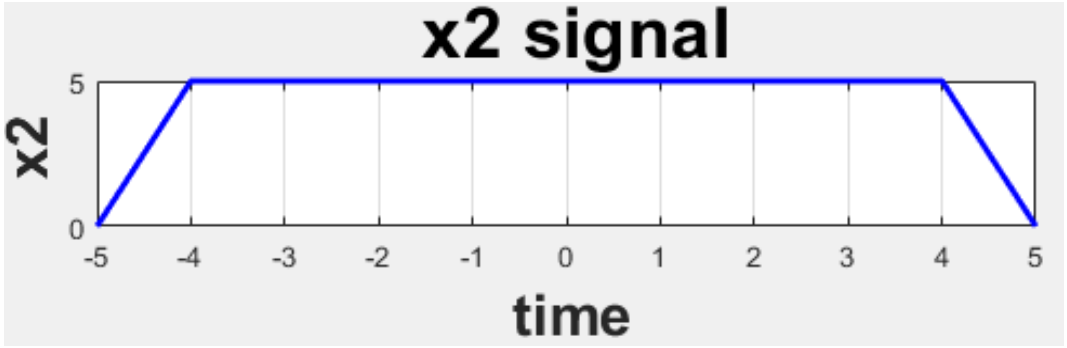
Outlines

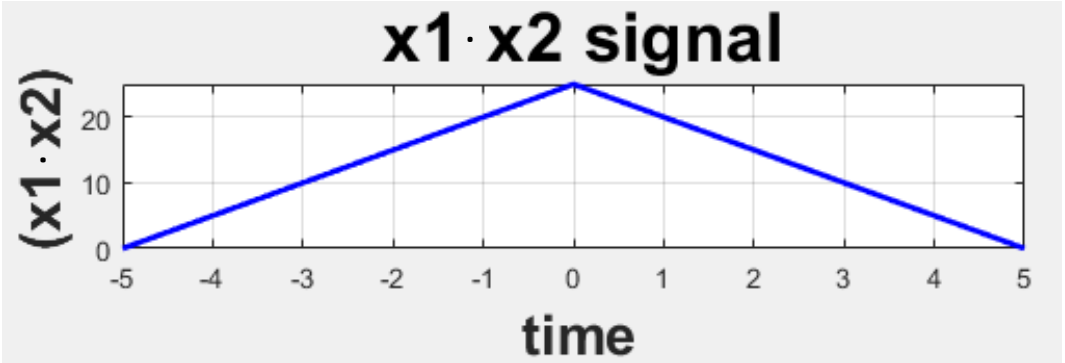
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Addition/multiplication Signals

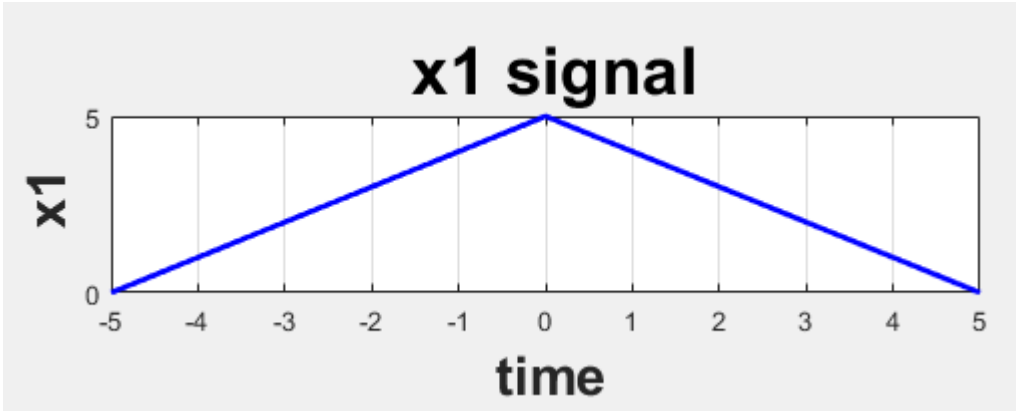
Two Signals are multiplier.

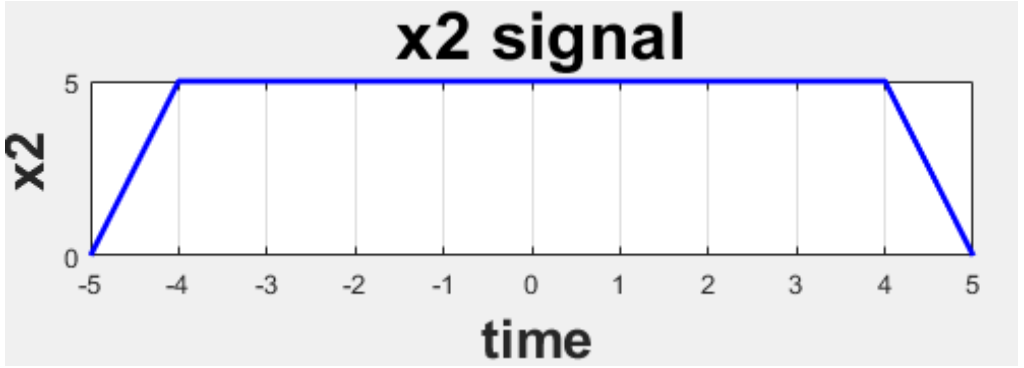


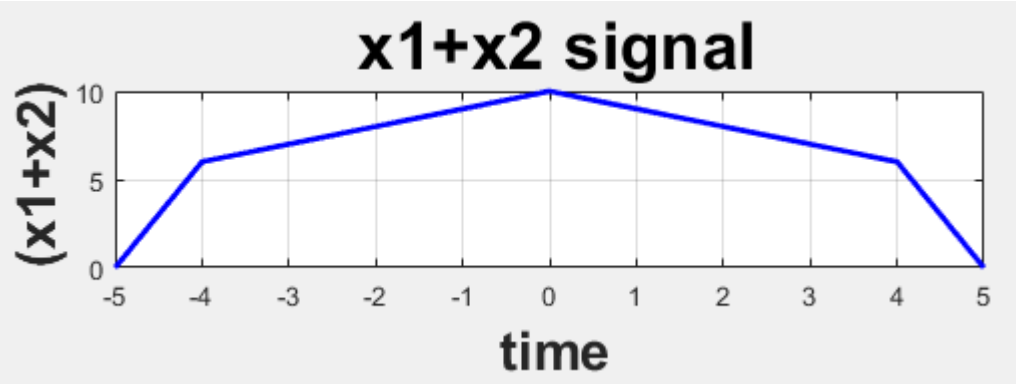




Two Signals are added.







Thanks!